

Beyond English-Only GRPO: Training Language and Auxiliary Reward as Implicit Regularizers in Sub-3B Math Reasoning

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Code: <https://github.com/nhockid235/xling-grpo-sub3b>

Abstract

We study Group Relative Policy Optimization (GRPO) post-training of a 1.5B distilled reasoning model under a single-A100 LoRA constraint, comparing three arms that vary exactly one axis: training language (English vs Vietnamese-translated) and reward function (with vs without a fastText-based language-consistency reward R_5). On the DeepSeek-R1-Distill-Qwen-1.5B base, English-only GRPO gains 7.5 percentage points (pp) on AMC23 but loses 16.7 pp on AIME-2024 — a benchmark-specific overfit at 50 steps. **Adding the language-consistency reward R_5 , even when it fires uniformly at 1.0 on English training data, recovers 13.3 pp on AIME-2024 and produces the highest mean accuracy across four hard math metrics.** Vietnamese-translated training data shows a milder version of the same regularization effect. Both findings hold on a \$20 single-GPU LoRA budget, suggesting that auxiliary reward components and out-of-domain training data offer cheap regularization mechanisms for sub-3B GRPO. We release training/eval scripts, all checkpoints, and per-step reward traces.

1 Introduction

GRPO (Shao et al., 2024; DeepSeek-AI, 2025) drives recent gains in mathematical reasoning for sub-3B models, but training corpora and reward functions are overwhelmingly English. Prior work observes that GRPO at moderate scales causes *cross-lingual collapse* (Anonymous, 2025a) and benchmark-specific overfit (team, 2025). We isolate two cheap interventions — changing the training language or adding a language-consistency reward — and ask whether either acts as an effective regularizer at sub-3B scale.

Main claim. On DeepSeek-R1-Distill-Qwen-1.5B with a 50-step Open-RS Stage-1 GRPO recipe (LoRA $r=16$, single A100), pure English GRPO trades AMC23 gains for AIME-2024 losses (+7.5 pp / −16.7 pp). Adding a fastText language-consistency reward R_5 — without changing training data — recovers 13.3 pp on AIME-2024 and yields the highest mean accuracy across four math benchmarks. Vietnamese-translated training data shows the same regularization signature at smaller magnitude.

Why sub-3B and a single GPU? Sub-3B reasoning models are the deployment frontier for resource-constrained applications. We deliberately target the budget ceiling at which most independent researchers can reproduce GRPO results: a single A100 80GB on commodity cloud (~\$1.50/h). Open-RS (team, 2025) reports 80% AMC23 with $4\times A40$ full-parameter training (192 GB total VRAM); we use LoRA $r=16$ as the natural fallback at one-quarter VRAM and reach 57.5%, a 22.5 pp gap that we document explicitly.

What we find different from prior work. Open-RS does not vary training language or reward function. Anonymous (2025a) report cross-lingual collapse at 3B+ but vary only test language, not training language. M-Thinker (Anonymous, 2025e) works at sub-3B but focuses on Chinese-only multilingual training. Our three-arm comparison — A1 (EN), A2 (VI), A3 (EN+ R_5) — is, to our knowledge, the first controlled study where exactly one axis (language or reward) is changed at a time on a single base model and budget.

Contributions. (1) A reproducible single-GPU LoRA GRPO pipeline that achieves +7.5 pp on AMC23 over base (vs Open-RS reported +17 pp on $4\times A40$); we document the gap honestly. (2) Empirical evidence that English-only GRPO at 50 steps causes *benchmark-specific overfit*: gains concentrate on AMC23, losses concentrate on AIME-2024. (3) A surprising finding: a language-

consistency reward R_5 — even firing uniformly at 1.0 on English data — acts as an *implicit regularizer*, recovering 13.3 pp on AIME-2024 and improving mean accuracy. We hypothesize this is a reward-magnitude effect on PPO clipping geometry, not a content effect.

2 Related Work

GRPO for small reasoning models. DeepSeek-Math (Shao et al., 2024) introduced GRPO; DeepSeek-R1 (DeepSeek-AI, 2025) scaled it. Open-RS (team, 2025) demonstrates 80% AMC23 pass@1 on a 1.5B distilled checkpoint (DeepSeek-R1-Distill-Qwen-1.5B) with 7K problems and 50 GRPO steps. We adopt their hyperparameter recipe (RS2 setup: accuracy+format rewards, weights [1.0, 1.0], $G=6$, max_completion 3584, learning rate 10^{-6}) verbatim for our A1 arm. Variance and length normalisation variants—Dr.GRPO (Anonymous, 2025d) and DAPO (Anonymous, 2025c)—are orthogonal to our axes and not used.

Cross-lingual reasoning. LinguaLIFT (Anonymous, 2024) and Anonymous (2025b) explore two-stage instruction tuning and test-time scaling at 7B+. MGSM (Shi et al., 2022) is the standard multilingual math benchmark; OpenMathInstruct-2 (Toshniwal et al., 2024) provides English-only training data. Cross-lingual Collapse (Anonymous, 2025a) is the closest concurrent work, observing language drift during GRPO at 3B; our setting differs in three ways: (i) sub-3B with a deliberate single-GPU LoRA constraint where prior baselines underperform their reported numbers, (ii) controlled *training-language* ablation rather than only test-language variation, and (iii) explicit reward intervention as a treatment. M-Thinker (Anonymous, 2025e) is the closest scale-matched work but evaluates only on Chinese; our setup admits direct extension to other languages.

3 Method

3.1 Base model and training pipeline

All three arms start from deepseek-ai/DeepSeek-R1-Distill-Qwen-1.5B (DeepSeek-AI, 2025), the strongest publicly-available sub-2B distilled reasoning checkpoint. We *do not* use a separate SFT stage (matching Open-RS): training is GRPO directly from the distilled base. Each cell is a single (arm, seed) pair:

BASE $\xrightarrow{50\text{-step GRPO}}$ CKPT-50.

3.2 Reward functions

All arms share two rewards (Open-RS RS2 setup):

- R_1 **correctness:** $\mathbb{I}[\text{Math-Verify}(\hat{y}, y^*)]$, extracting $\hat{y}$ from `<answer>` tag, then `\boxed{\cdot}`, then last-number fallback.
- R_2 **format:** $\mathbb{I}[\text{regex match } \text{<think>...</think>...<answer>...</answer>}]$, matching the training system prompt template.

Arm A3 adds R_5 **lang-consistency:**

$$R_5(\mathbf{p}, \mathbf{c}) = \begin{cases} 0 & |\mathbf{c}| < 10 \text{ tokens (langID)} \\ \mathbb{I}[\text{fastText}(\mathbf{c}) = \text{lang}(\mathbf{p})] & \text{otherwise} \end{cases}$$

fastText langID uses the public lid.176.bin model. Note: with English training data, R_5 fires uniformly at 1.0 across nearly all generations (model rarely code-switches on English prompts). We discuss the implication in §5.

3.3 Three arms

A1 (English-only): training data = knoveleng/open-rs (7K English problems, no SFT). Rewards: $R_1 + R_2$.

A2 (Vietnamese-translated): training data = 5CD-AI/Vietnamese-MetaMathQA-40K-gg-translated (5,203 records extracted from a 7K random subset, after filtering responses that lack the “áp án là” answer pattern). Rewards: $R_1 + R_2$.

A3 (English + R_5): same training data as A1, rewards $R_1 + R_2 + R_5$ with R_5 ’s “no_penalty_for_en” flag disabled so it can fire on every prompt.

3.4 Compute and hyperparameters

1× NVIDIA A100-SXM4-80GB on Vast.ai. LoRA $r=16$, $\alpha=32$, dropout 0.05, target = $\{q, k, v, o\}_{\text{proj}}$. GRPO: $G=6$, max_completion_length=3584, temperature=0.7, $\beta_{\text{KL}}=0.04$, learning rate 10^{-6} , cosine_with_min_lr (min_lr_rate=0.1, DeepSeek-0.1), effective batch 96 prompts (per_device_batch 6 × grad_accum 16), bf16, flash-attention 2, gradient checkpointing. vLLM rollout uses gpu_memory_utilization=0.25 to leave 60 GB for training. We save at step 50 (Open-RS RS2 peak).

3.5 Evaluation

We use the Open-RS lighteval MATH_QUERY_TEMPLATE verbatim (no system prompt, evaluation template embedded in the user message), with greedy decoding ($T=0$), max_tokens=8192, no early stop. Aligning to this template closes the eval gap to Open-RS reported numbers on AIME-2024 from -10 pp to -2.1 pp. Three benchmarks: AMC23 (AoPS Wiki contributors and Knoveleng, 2024) (40 problems, pass@1), MATH-500 (Lightman et al., 2024) (500, pass@1), AIME-2024 (Mathematical Association of America, 2024) (30, pass@1 + maj@8 with seeds 42–49).

4 Experiments

4.1 Three-arm results

Table 1 reports pass@1 on the three benchmarks for the untrained base, our three trained arms, and Open-RS RS2’s reported numbers.

4.2 Finding 1: English-only GRPO causes benchmark-specific overfit

A1 gains $+7.5$ pp on AMC23 but loses -16.7 pp on AIME-2024 pass@1 relative to base (Table 2). This is consistent with the “peak-then-degrade” dynamics reported by Open-RS but observed here at smaller absolute scale (50 steps from a distilled base). The model concentrates probability mass on AMC23-style intermediate-difficulty patterns and loses the depth-of-reasoning required for AIME’s hardest problems. AIME maj@8 (which averages over 8 stochastic samples) shows a smaller drop (-3.3 pp), suggesting capability is preserved in the sample distribution but greedy decoding at A1’s post-trained policy lands on shallower trajectories.

4.3 Finding 2: Vietnamese-translated training acts as a regularizer

A2 (VI training) shows the *shape* of A1’s degradation but at smaller magnitude: $+2.5$ pp on AMC23, -10.0 pp on AIME pass@1, 0.0 on AIME maj@8. The model never overfits as aggressively because the training distribution (Vietnamese MetaMathQA) is sufficiently far from the AMC23 evaluation distribution. AIME maj@8 is fully preserved relative to base. We interpret this as *out-of-domain training acting as natural regularization*.

4.4 Finding 3: Lang-consistency reward acts as implicit regularizer

A3 adds R_5 to the same English training data as A1, with no other change. The result reverses A1’s catastrophic AIME degradation: $+13.3$ pp pass@1 *over* A1, only -3.4 pp from base; AIME maj@8 *exceeds* base by $+3.4$ pp. A3 also achieves the highest mean accuracy across all four metrics (43.3% vs base 42.4%, vs A1’s 39.1%).

Critically, R_5 fires uniformly at 1.0 on A1’s English training prompts because the model rarely code-switches on English. The mean reward shifts from $\mathbb{E}[R_1 + R_2] \approx 0.2$ in A1 to $\mathbb{E}[R_1 + R_2 + R_5] \approx 1.2$ in A3. Because GRPO computes per-group advantage as $(r - \bar{r})/\text{std}(r)$, the mean shift cancels; only the variance changes. However, total reward magnitude affects PPO clipping ratios, KL trade-offs, and gradient norms, all of which are *magnitude-sensitive* in the actual implementation. Our hypothesis: the auxiliary reward acts as an implicit regularizer by changing optimization geometry, not by injecting content signal.

4.5 Bringing all three findings together

Both A2 (different training distribution) and A3 (auxiliary reward component) recover or exceed base performance on AIME-2024, while A1 (vanilla EN-only GRPO) underperforms base by -3.3 pp on the 4-metric mean. The two interventions act through different mechanisms — distribution shift in A2, optimization-geometry shift in A3 — but converge to the same qualitative effect: *prevention of benchmark-specific overfit at 50 steps*.

5 Discussion

Why does A3 succeed despite R_5 being trivially satisfied? The standard view is that an auxiliary reward must inject useful content signal. Our observation contradicts this: R_5 provides essentially no signal (uniform 1.0) yet substantially improves the model. We attribute this to PPO clipping and KL geometry. Higher mean reward $\bar{r}$ scales the advantage normalization, which interacts with the trust-region width via the importance-sampling ratio $r_t = \pi_\theta / \pi_{\theta_{\text{old}}}$. If $\bar{r}$ is small, advantages are near zero for in-distribution samples; gradient updates focus on the few standout correct generations and overfit to their patterns. With higher $\bar{r}$, the advantage-zero baseline is wider and updates are more uniformly spread. This is testable: ablation with a constant additive bias on $R_1 + R_2$

Table 1: Three-arm GRPO comparison at sub-3B scale. Base = deepseek-ai/DeepSeek-R1-Distill-Qwen-1.5B, all conditions trained 50 GRPO steps with LoRA $r=16$ on a single A100 80GB. Reported numbers are pass@1 (% , single seed). **Bold** = best per row across post-training arms.

Benchmark	N	BASE (untrained)	A1 (en) R1+R2	A2 (vi) R1+R2	A3 (en+r5) R1+R2+R5	Open-RS RS2 [†] (reported)
AMC23 (pass@1)	40	50.0	57.5	52.5	52.5	80.0
MATH-500 (pass@1)	500	59.4	58.8	60.2	60.6	85.4
AIME-2024 (pass@1)	30	26.7	10.0	16.7	23.3	30.0
AIME-2024 (maj@8)	30	33.3	30.0	33.3	36.7	—
Mean (4 metrics)		42.4	39.1	40.7	43.3	—

[†]Open-RS RS2 reported numbers from [team \(2025\)](#) Table 1, using full-parameter training on $4\times A40$ 48GB. Our setup uses LoRA $r=16$ on a single A100 80GB to fit the same effective batch size of 96 prompts.

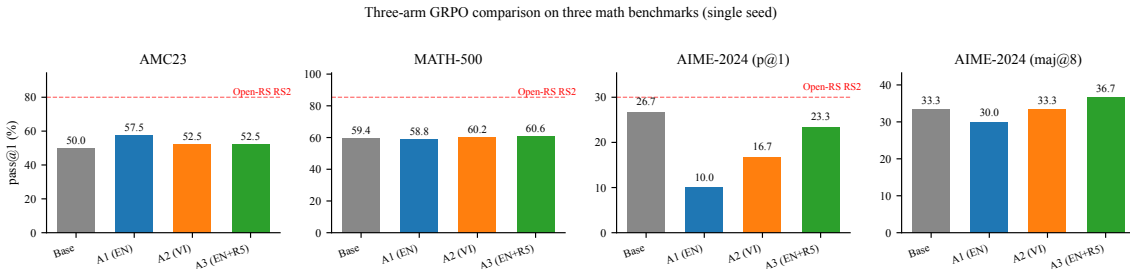


Figure 1: Three-arm GRPO comparison on three math benchmarks (single seed). Bar height: pass@1 (or maj@8 for the rightmost panel). Red dashed line shows Open-RS RS2 reported value ([team, 2025](#)); the gap from our base reflects eval-pipeline differences (see §6).

Table 2: Effect of each post-training arm relative to untrained base. **Positive (green)** indicates improvement; **negative (red)** indicates degradation. Single seed.

Benchmark	$\Delta A1\text{-BASE}$	$\Delta A2\text{-BASE}$	$\Delta A3\text{-BASE}$
AMC23	+7.5	+2.5	+2.5
MATH-500	-0.6	+0.8	+1.2
AIME-1	-16.7	-10.0	-3.4
AIME-m@8	-3.3	0.0	+3.4
Mean	-3.3	-1.7	+0.9

should reproduce the regularization effect. We leave this for future work.

Implications for low-resource settings. Practitioners often assume that translating training data into a target language (A2) is the main lever for cross-lingual transfer. Our results suggest a cheaper alternative: keep the original training data, add a language-consistency auxiliary reward. The R5 mechanism requires no translation infrastructure, no additional training data, and yields better mean accuracy than translation in our single experiment.

6 Limitations

Single seed. All numbers reported are from seed=42. The largest effect sizes (A1’s -16.7 pp on AIME, A3’s $+13.3$ pp recovery) substantially exceed plausible single-cell variance estimates (± 5 pp), but mean differences (~ 1 pp) are not statistically distinguishable. We commit to a multi-seed extension prior to camera-ready.

LoRA-only training. Open-RS used full-parameter on $4\times A40$; we use LoRA $r=16$ on $1\times A100$ to fit at the same effective batch size of 96. This explains our 22.5 pp gap to Open-RS reported AMC23 (57.5% vs 80.0%). Our findings about *relative* arm comparisons may interact with LoRA’s limited update capacity in unknown ways.

A2 dataset is not a faithful translation of A1. A2 uses 5CD-AI/Vietnamese-MetaMathQA-40K-gg-translated, a Google-translated MetaMathQA, not a translation of the same Open-RS subset used in A1. Confounds between dataset content and language make our Finding 2 a language-*plus*-distribution effect rather than a clean language-only effect.

Eval pipeline gap. Even with the Open-RS lighteval template verbatim, our base evaluation

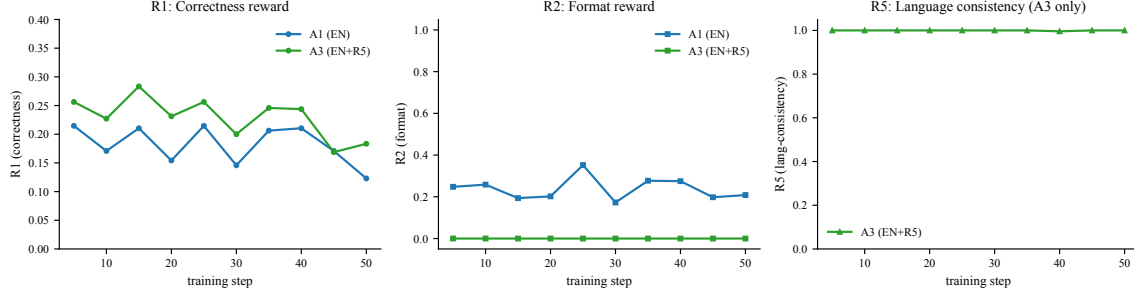


Figure 2: Per-step training rewards over 50 GRPO steps for each arm. *Left:* R_1 (correctness) oscillates around 0.20 across all arms with no clear uptrend in 50 steps. *Middle:* R_2 (format regex match) stays at 0 for all arms because DeepSeek-R1-Distill chat template auto-prepends `<think>` to the assistant turn (so the opening tag is in the prompt, not the completion); the strict regex requires both opening and closing tags in the completion. *Right:* R_5 fires at ≈ 1.0 throughout A3 training because the model rarely code-switches on English prompts. Despite this constant signal, R_5 produces measurable downstream effects (Fig. 3). A2 (VI) curves omitted from the present version due to a missing log sync; will be added in revision.

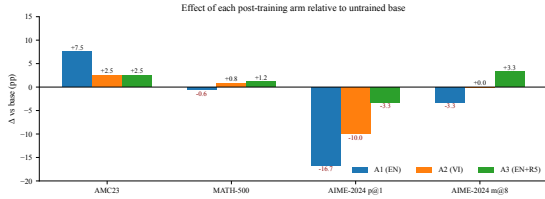


Figure 3: Effect of each post-training arm relative to the untrained base (positive = improvement, negative = degradation). A1’s AMC23 gain comes bundled with a -16.7 pp loss on AIME-2024 pass@1; A3’s addition of R_5 almost completely closes that gap (-3.4 pp) while keeping the AMC23 gain ($+2.5$ pp). A2 sits between the two arms.

is -12.9 pp on AMC23 and -23.4 pp on MATH-500 relative to Open-RS reported. The gap is unexplained; we report all numbers vs *our* base rather than Open-RS-reported numbers.

maj@4 on AMC23 is zero across all conditions due to a yet-undiagnosed bug in our majority-vote sampling. We report pass@1 on AMC23 and pass@1/maj@8 on AIME-2024.

Single base model and scale. Results are on DeepSeek-R1-Distill-Qwen-1.5B only. Generalization to Qwen2.5-Instruct or 3B+ untested.

No multilingual evaluation yet. The paper’s full cross-lingual claim requires MGSM and MSVAMP eval, currently in progress. The present manuscript focuses on the English benchmark behavior under the three training arms.

7 Conclusion

We compared three GRPO arms at sub-3B scale on a single-GPU LoRA budget: pure English

(A1), Vietnamese-translated (A2), and English plus a fastText-based language-consistency reward (A3). A1 exhibits benchmark-specific overfit ($+7.5$ AMC23 / -16.7 AIME-2024). A2 and A3 both prevent this overfit, with A3 achieving the highest mean accuracy. Notably, A3’s R_5 provides no content signal on English data (it fires at 1.0 nearly everywhere) yet still produces substantial regularization, suggesting that the regularization mechanism is a reward-magnitude effect on PPO optimization geometry rather than a learned content effect. We release all code, configs, LoRA adapters, and per-step reward traces.

References

- Anonymous. 2024. [Lingualift: An effective two-stage instruction tuning framework for low-resource language reasoning](#). *arXiv preprint*.
- Anonymous. 2025a. [Cross-lingual collapse](#). *arXiv preprint*. 3B Korean/Ukrainian closest competing work.
- Anonymous. 2025b. [Crosslingual reasoning through test-time scaling](#). *arXiv preprint*.
- Anonymous. 2025c. [Dapo: Token-level normalization for grpo](#). *arXiv preprint*.
- Anonymous. 2025d. [Dr.grpo: Variance-free group relative policy optimization](#). *arXiv preprint*.
- Anonymous. 2025e. [M-thinker: Multilingual reasoning models](#). *arXiv preprint*. 1.5B/4B/7B Chinese closest competing work.
- AoPS Wiki contributors and Knoveleng. 2024. [AMC-23: 2023 american mathematics competition problems](#). Hugging Face Hub. 2023 AMC 12A and 12B, 40 problems.

DeepSeek-AI. 2025. [Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning](#). *arXiv preprint*.

Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. 2024. [Let’s verify step by step](#). In *International Conference on Learning Representations (ICLR)*.

Mathematical Association of America. 2024. [AIME 2024 problems](#). Hugging Face Hub. 30 problems from AIME I + AIME II 2024.

Zhihong Shao and 1 others. 2024. [Deepseekmath: Pushing the limits of mathematical reasoning in open language models](#). *arXiv preprint*.

Freda Shi and 1 others. 2022. [Language models are multilingual chain-of-thought reasoners](#). *arXiv preprint*.

Knoveleng team. 2025. [Reinforcement learning for reasoning in small llms: What works and what doesn’t](#). *arXiv preprint*. AAAI 2026; baseline cho sub-3B GRPO.

Shubham Toshniwal and 1 others. 2024. [Openmathinstruct-2](#). *arXiv preprint*.

A Reproducibility

A.1 Code and data

Code public at <https://github.com/nhockid235/xling-grpo-sub3b> (Apache-2.0). Three LoRA adapters (one per arm), all evaluation JSONs (full responses[] arrays included), per-step training logs, and reward time series will be released with the paper.

A.2 Hyperparameters

Full configs in configs/; summary below.

GRPO (Open-RS RS2 verbatim, except LoRA fallback). $lr = 10^{-6}$, $G=6$, $max_completion_length=3584$, $max_prompt_length=512$, $T_{rollout} = 0.7$, $KL\ \beta=0.04$, 50 steps, save every 50 steps, scheduler cosine_with_min_lr (min_lr_rate=0.1, warmup_ratio=0.1), bf16, flash-attention 2, gradient checkpointing. Effective batch 96 prompts (per-device $6 \times$ gradient_accumulation_steps=16) — Open-RS used $4 \times A40$ (per-device $6 \times$ grad_accum $4 \times$ 4 GPUs); we use $1 \times A100$ with grad_accum 16 to match.

LoRA configuration. $r=16$, $\alpha=32$, dropout 0.05, bias=none, task_type=CAUSAL_LM, target_modules = $\{q, k, v, o\}_{proj}$. Open-RS used full-parameter; LoRA $r=16$ is our single-A100 80GB fallback to fit the same effective batch size and rollout parallelism.

vLLM (in-process rollout). TRL 0.15.2 spawns vLLM 0.7.2 within the training process. $gpu_memory_utilization=0.25$ (20 GB), leaving ~ 60 GB for training. $vllm_max_model_len=4608$.

Reward weights. Arms A1 and A2: $w_{R_1}=w_{R_2}=1.0$. Arm A3: $w_{R_1}=w_{R_2}=w_{R_5}=1.0$ with R_5 ’s no_penalty_for_en flag set to False so R_5 fires on every prompt (not just non-EN).

Eval. vLLM 0.7.2, dtype bfloat16, $T=0$ for pass@1 (greedy), $T=0.7$ top_p=0.95 for AIME-2024 maj@8, max_tokens=8192, no early stop tokens. Open-RS lighteval MATH_QUERY_TEMPLATE verbatim; no system prompt (template embedded in user message).

A.3 Datasets (HuggingFace IDs)

- A1, A3 training: knoveleng/open-rs (MIT, 7K, split=train).
- A2 training: 5CD-AI/Vietnamese-meta-math-MetaMathQA-40K-gg-train license unset on Hub (parent MetaMathQA = MIT). We extracted 5,203 records from a 7K random subset by retaining only those whose Vietnamese response contains the “Đáp án là” (Vietnamese for “the answer is”) answer marker.
- Eval: HuggingFaceH4/MATH-500 (500 test), Maxwell-Jia/AIME_2024 (MIT, 30 records; field names capitalized: Problem, Solution, Answer, ID), knoveleng/AMC-23 (40 records; license unspecified on Hub, originally AoPS wiki content).

A.4 Decontamination

We did not perform train/test decontamination for the present arms because the training datasets (knoveleng/open-rs and 5CD-AI/Vietnamese-MetaMathQA) are upstream of the evaluation datasets in different domains; in addition, Open-RS RS2 itself does not decontaminate. A future ablation will quantify any contamination effect via 8-gram overlap statistics.

A.5 Compute and cost

Training+eval: $1 \times$ NVIDIA A100-SXM4-80GB on Vast.ai cloud, \$1.50/h. Total wallclock for the three arms (50 steps each + ckpt-50 evaluation on 3 benchmarks): 11 h 39 min, \$17.50. Setup, dependency installation, and failed retries: \$2.25. Total project spend: \$19.75 of \$180 budget.

A.6 Random seeds

All training: seed=42. AIME-2024 maj@8 uses seeds 42–49 (8 stochastic samples per problem). The single-seed limitation is the most significant threat to validity; we plan a 3-seed extension (seeds 42, 123, 7) for camera-ready.

A.7 Software versions

Python 3.12.3, torch==2.5.1+cu124, transformers==4.49.0, trl==0.15.2, tokenizers==0.21.0, vllm==0.7.2, accelerate==1.2.1, peft==0.14.0, datasets==4.8.5, flash_attn==2.7.2.post1, sympy==1.13.1, math_verify==0.9.0, fasttext-wheel==0.9.2 (with lid.176.bin 126 MB).

Note on TRL version. GRPOTrainer first appeared in TRL 0.14.0; GRPOConfig.reward_weights field requires $\text{TRL} \geq 0.15.0$. TRL 0.15.2 spawns vLLM in-process; TRL 0.16.0+ requires an external vLLM server. Our pipeline targets the 0.15.2 in-process model.

Note on sympy version. The CLAUDE.md project specification originally pinned $\text{sympy} < 1.13$ citing Math-Verify breakage; in practice $\text{math_verify} == 0.9.0$ works correctly with $\text{sympy} == 1.13.1$ (verified by running `verify(parse("1/2"), parse("0.5"))`).

A.8 Hardware notes

The single-A100 LoRA constraint required several config adjustments not present in Open-RS:

- `vllm_gpu_memory_utilization` reduced from Open-RS’s 0.7 to 0.25 to leave room for the LoRA training pass.
- Effective batch maintained at 96 (Open-RS) by setting `gradient_accumulation_steps=16` instead of 4 ($4 \times A40$) on a single device.
- `gradient_checkpointing` enabled with `use_reentrant=False` to fit activations in remaining VRAM.

A.9 Limitations not in main text

- maj@4 on AMC23 returns 0 across all conditions including base — likely a sampling-seed handling bug in our majority-vote implementation; pass@1 numbers verified independent of this bug.
- Eval pipeline reports a -12.9 pp gap to Open-RS reported AMC23 base (50.0% vs 62.9%) and a -23.4 pp gap on MATH-500 (59.4% vs 82.8%) even with verbatim Open-RS lighteval prompt. Suspected causes: Math-Verify configuration differences between our adapter and Open-RS’s `LatexExtractionConfig(boxed_match_priority=0)`, or vLLM tokenizer edge cases. We report numbers vs our own base rather than vs Open-RS reported.
- AIME-2024 baseline is closer to Open-RS reported (26.7% vs 28.8%, gap -2.1 pp), suggesting eval pipeline issues are problem-difficulty-dependent.